

YU YUAN

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RESEARCH INTERESTS

Visual Understanding and Generation, Physically-Consistent Image/Video Synthesis, World Models, Computational Photography, Diffusion Models, Generative Modeling, Image/Video Enhancement.

EDUCATION

Purdue University

PhD Student in Electrical and Computer Engineering Advisor: Stanley H. Chan

West Lafayette, IN, US

08/2023 – 05/2027 (expected)

Shanghai Jiao Tong University

MS in Aeronautical and Astronautical Science and Technology

Shanghai, China

09/2020 – 06/2023

Shanghai Jiao Tong University

BEng in Aeronautics and Astronautics Engineering

Shanghai, China

09/2016 – 06/2020

Minor Degree in Administration Management

10/2018 – 06/2020

PROJECTS EXPERIENCE

Purdue University

Research Assistant (Advisor: Prof. Stanley H. Chan)

West Lafayette, IN

08/2023 – Present

- **NewtonGen (Physics-Consistent Text-to-Video Generation):**

- * **Developed** a physics-consistent and controllable text-to-video framework that explicitly incorporates learnable dynamics.
- * **Innovated** Neural Newtonian Dynamics, which models different dynamics via unified neural ODEs.

- **Generative Photography (Scene-Consistent Camera-Controlled Text-to-Image Generation):**

- * **Introduced** a new text-to-image generation paradigm for photography with an understanding of camera physics.
- * **Produced** significantly more scene-consistent photorealistic images than SOTA models (Stable Diffusion 3, FLUX).

- **High Dynamic Range (HDR) Imaging:**

- * **Proposed** a novel high dynamic range (HDR) imaging framework capable of handling a flexible number of causal inputs.
- * **Utilized** side information and attention mechanisms to effectively suppress artifacts during the fusion process.

- **Astrophotography Turbulence Mitigation with Generative Models:**

- * **Established** the first large-scale dataset for astronomical image turbulence restoration, comprising over 135K images.
- * **Presented** a novel architecture integrating a generative prior with a restoration model for structural preservation.

PUBLICATIONS

Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, Stanley H. Chan.

Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis.

CVPR 2025 [Highlight & Demo]

Yu Yuan, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, Bole Ma, Stanley H. Chan.

NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics.

arXiv 2025 [Under Review]

Yu Yuan, Yiheng Chi, Xingguang Zhang, Stanley H. Chan.

iHDR: Iterative HDR Imaging with Arbitrary Number of Exposures.

ICIP 2025

Xingguang Zhang, Nicholas Chimitt, Xijun Wang, **Yu Yuan**, Stanley H. Chan.
Learning Phase Distortion with Selective State Space Models for Video Turbulence Mitigation.
CVPR 2025 [Highlight]

Lanqing Guo*, Xijun Wang*, Minchul Kim*, **Yu Yuan**, Wes Robbins, Xingguang Zhang, Stanley H. Chan, Zhangyang Wang, Xiaoming Liu. ***Beyond Clean Pixels: Interdisciplinary Lessons from Turbulent Long-Range Face Recognition.***
2025 [Under Review]

Joonyeoup Kim, **Yu Yuan**, Xingguang Zhang, Xijun Wang, Stanley H. Chan.
Astrophotography Turbulence Mitigation via Generative Models.
ICIP 2025

Lindong Wang, Hongya Tuo, **Yu Yuan**, Henry Leung, Zhongliang Jing.
RCMixer: Radar-Camera Fusion Based on Vision Transformer for Robust Object Detection.
JVCJ 2024

Yu Yuan, Jiaqi Wu, Lindong Wang, Zhongliang Jing, Henry Leung, Shuyuan Zhu, Han Pan.
Learning to Kindle the Starlight.
arXiv 2022

Yu Yuan, Jiaqi Wu, Zhongliang Jing, Henry Leung, Han Pan.
Multimodal Image Fusion based on Hybrid CNN-Transformer and Non-local Cross-modal Attention.
arXiv 2022

PATENT

A trajectory solving and alignment method based on inertial measurement unit and ultra-short baseline positioning sensors for autonomous underwater robots. China Patent. CN 114993313 B.

AWARDS & COMPETITIONS

Ivy League Scholarship for Exceptional Students (Innovation)	Top 1%
Outstanding Graduate of Shanghai Jiao Tong University	Top 15%
Shanghai International Creators Competition: Special Competition of Drones	First Prize
DELL AI for Social Innovation Competition: AI/ADAS of Intelligent Car	National Second Prize

OTHER ACTIVITIES

Teaching Assistant , Automatic Control Theory, Shanghai Jiao Tong University	2021
Workshop Co-organizer , WACV GAIP 2026	
Reviewer , CVPR, CVPRW, ICCV, WACV, ICIP, ICASSP	
IEEE Graduate Student Member	

SKILLS

Languages: English (Fluent), Chinese (Native)
Programming: Python, PyTorch, HTML, JavaScript, Matlab, C/C++ (Basic)
Technologies: Diffusers, Transformers, Post-training, 3DGS, CUDA, OpenCV, NumPy, Git, LaTeX, Linux, Slurm, Markdown, Gradio, NVIDIA Jetson, Adobe Photoshop, Adobe Premiere
Others: Photography, Fishing, Drones, Rowing